Double-Phase Hologram Steganography using Keyed Gabor Splatting Representations

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Steganography is a well-established technique for hiding information in images and video, while computer-generated holograms, the content that emerging holographic displays rely on, remain largely unexplored as steganographic carriers. We introduce a Quantization Index Modulation (QIM)-based information-hiding method using Gabor Splatting (GS) representations of Double-Phase Holograms (DPHs). A DPH encodes complementary checkerboard phase maps, which we unpack and fit with explicit 2D Gabor primitives. We test how changes to different Gabor primitive parameters affect holographic reconstruction quality and select the parameter that carries the payload. Keyed QIM then embeds bits into the selected parameter, and the decoder recovers the payload from the saved representation and shared key. At a fixed 100-bit payload, GS produces the highest-quality holographically reconstructed images, exceeding the leading baseline by 2.23 dB in PSNR over 200 DPHs. Across payloads of 25–1000 bits, the distortion added by QIM remains below the Gabor fitting error, so reconstruction quality is limited by the representation rather than by embedding.

1. INTRODUCTION

Steganography conceals secret information within a cover medium to make the communication itself difficult to detect. Steganographic methods have long used images and video as visual carriers by modifying pixel values or coefficients derived from them. Holographic displays are an emerging technology for reconstructing three-dimensional scenes from computer-generated holograms [1]. Like images and video, holograms must be stored and transmitted as display content. Yet they remain largely unexplored as steganographic carriers. Existing holographic information-hiding methods either treat the hologram as the hidden content or modify the hologram itself. Some methods encode a secret image as a hologram and embed the resulting hologram data in a conventional host image [2, 3]. Others embed watermarks directly in hologram values or transform coefficients, incorporate auxiliary data during hologram generation, or use holograms for authentication and encryption [4–6]. In all of these methods, the payload resides in image pixels, hologram samples, or transform coefficients, or is inserted during hologram generation. They do not embed payloads into the parameters of a saved explicit representation after hologram generation or recover them from those parameters. For an extensive review of holographic information hiding and parameter-domain methods, see Supplement 1, Sec. S1.

We introduce a keyed QIM method for hiding information in readily available DPHs using **GS** representations [7]. Following Fan et al. [8], we unpack each DPH into its high-value and low-value phase components. Yet, we replace their 2D Gaussian primitives with 2D Gabor primitives to create a compressed, streamable representation of the DPH. We compare candidate parameter perturbations in the **GS** representation by utilizing the visual quality of the holographic reconstructions and select effective log amplitudes. Upon this comparison, we embed bits into the selected primitives using keyed QIM. For recovery, the decoder reads the saved primitive state rather than the rendered DPH and uses the shared key to regenerate carrier selection, bit order, and dithers. For evaluation, we compare our method against **GaussianImage** [9], an adapted **INRSteg** model [10], and the image-domain **RoSteALS** method [11] at a fixed 100-bit payload on 200 images. Our method produces the highest-quality holographically reconstructed images among the evaluated methods, exceeding **RoSteALS**, the strongest baseline, by 2.23 dB in PSNR. Across payloads ranging from 25 to 1000 bits, the distortion introduced by QIM remains below the Gabor fitting error. Given the results, **GS** establishes itself as a practical parameter-domain carrier for holography. Upon acceptance, our code will be available at GitHub: complight/holographic_steganography.

2. METHOD

Given an already generated DPH, we separate its checkerboard components, fit them with Gabor primitives, and embed the payload into selected primitive parameters using keyed QIM. Fig. 1 shows an overview of our method.

DPH and checkerboard decomposition. Let $\varphi \in [0, 2\pi)^{K\times H\times W}$ denote a K-channel physical DPH patch. Following Fan et al. [8], we split it into the high-value φ_{hi} and low-value φ_{lo} components. We fit each component φ_s, $s \in \{\mathrm{hi}, \mathrm{lo}\}$, independently with **GS** and, after fitting and embedding, interleave the rendered outputs at the original checkerboard locations to restore the DPH.

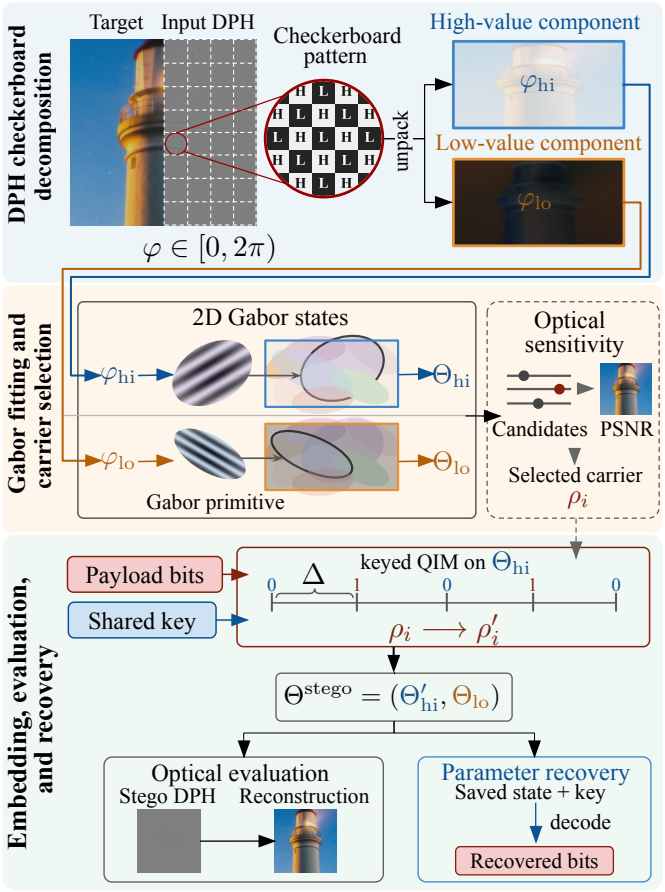


Fig. 1. Overview of the proposed pipeline. We unpack the DPH into high- and low-value components and fit with Gabor primitives. The dashed path denotes the one-time offline sensitivity analysis that selects effective log amplitude. Keyed QIM modifies selected high-value parameters, and the state supports holographic reconstruction and payload recovery.

GS representation. For each component $s \in \{\mathrm{hi}, \mathrm{lo}\}$, **GS** [7] represents φ_s with N_{Gab} primitives whose parameters form the state Θ_s. Primitive i carries a color vector $\mathbf{c}_{i,s} \in \mathbb{R}^K$ and an anisotropic Gaussian envelope $g_{i,s}^{\mathrm{Gab}}(\mathbf{x})$ defined by its center and covariance. The envelope is modulated by a set of atoms $\mathcal{K}_{i,s}$: the two largest-magnitude atoms selected from a 16-entry frequency bank. Each atom $k \in \mathcal{K}_{i,s}$ has coefficient $a_{ik,s}$, angular spatial frequency ω_{ik}, and oscillates along $t_{x,i,s}(\mathbf{x})$, the coordinate on the primitive's rotated local axis. Collecting the active primitive–atom pairs $\mathcal{C}_s = \{(i,k) : 1 \le i \le N_{\mathrm{Gab}}, k \in \mathcal{K}_{i,s}\}$ with their combined vector coefficients $\mathbf{b}_{ik,s} = a_{ik,s}\mathbf{c}_{i,s} \in \mathbb{R}^K$, the Gabor field is defined as:

$$\mathbf{F}_{\mathrm{Gab},s}(\mathbf{x}) = \sum_{(i,k)\in\mathcal{C}_s} \mathbf{b}_{ik,s}\, g_{i,s}^{\mathrm{Gab}}(\mathbf{x}) \cos(\omega_{ik} t_{x,i,s}(\mathbf{x})) \,. \quad (1)$$

We optimize $\mathbf{c}_{i,s}$ and $a_{ik,s}$ as separate parameters rather than their product $\mathbf{b}_{ik,s}$ directly. Let Ω denote the spatial domain of one unpacked component, $\ell \in \{1, \dots, K\}$ the phase-channel index, and $\widetilde{\varphi}_s = 2\pi \mathbf{F}_{\mathrm{Gab},s}$ the unconstrained phase output. Because phase is periodic, we train **GS** with the circular objective [12], averaged over $s \in \{\mathrm{hi}, \mathrm{lo}\}$, $\mathbf{x} \in \Omega$, and ℓ:

$$\mathcal{L}_{\mathrm{circ}} = \frac{1}{2|\Omega|K} \sum_{s,\mathbf{x},\ell} [1 - \cos(\widetilde{\varphi}_{s,\ell}(\mathbf{x}) - \varphi_{s,\ell}(\mathbf{x}))] \,. \quad (2)$$

This loss equals half the squared distance between unit phasors and is therefore invariant to integer multiples of 2π. For serialization and evaluation, the trained field is converted to physical phase and wrapped modulo 2π:

$$\widehat{\varphi}_{\mathrm{Gab},s} = 2\pi \,\mathrm{frac}\big(\mathbf{F}_{\mathrm{Gab},s}\big), \qquad \mathrm{frac}(v) = v - \lfloor v \rfloor. \quad (3)$$

For comparison, **GaussianImage** fits the same components with anisotropic Gaussian primitives without cosine modulation, using the sinusoidal DPH output mapping of Fan et al. [8].

Identifying parameters for payload embedding. Explicit representations expose several editable parameters, but changing different parameters can produce different phase and optical errors. We therefore hold the clean **GaussianImage** and **GS** fits fixed, perturb each parameter family independently, restore the complete DPH, and evaluate the resulting changes in phase and holographic reconstruction. This analysis selects covariance log-area in the low-value component for **GaussianImage** and effective log amplitude in the high-value component for **GS**.

Keyed one-branch QIM embedding and recovery. For **GS**, we embed one bit in each selected primitive of Θ_{hi} and leave Θ_{lo} unchanged. Suppressing the fixed high-value component index, let $\mathbf{B}_i = [\mathbf{b}_{ik}]_{k\in\mathcal{K}_i}$ denote the active coefficient block, with $\mathbf{a}_{i,\mathcal{K}_i} = [a_{ik}]_{k\in\mathcal{K}_i}$. We define $\|\cdot\|_{\mathrm{F}}$ as the Frobenius norm. Its effective log amplitude is

$$\rho_i = \log\|\mathbf{B}_i\|_{\mathrm{F}} = \log(\|\mathbf{c}_i\|_2 \|\mathbf{a}_{i,\mathcal{K}_i}\|_2) \,. \quad (4)$$

Only finite primitives with valid nonzero effective amplitude are eligible carriers. For payload bit $m_i \in \{0,1\}$, lattice spacing Δ, and keyed centered dither $d_i \in [-\Delta/2, \Delta/2)$, keyed QIM [13] restricts the integer lattice index n to even values when $m_i = 0$ and to odd values when $m_i = 1$, selects the nearest such index q_i, and replaces ρ_i with the embedded log amplitude ρ_i'.

$$q_i = \operatorname*{argmin}_n |\rho_i - d_i - n\Delta| \,, \qquad \rho_i' = d_i + q_i\Delta. \quad (5)$$

We realize this update as $\mathbf{B}_i' = \exp(\rho_i' - \rho_i)\mathbf{B}_i$, scaling all active wave coefficients by the same factor, and the decoder recovers m_i as the parity of the nearest lattice index. The reported experiments use $\Delta = \log(1.0025)$, i.e., a 0.25% amplitude step, selected on a separate calibration subset. Because embedding alters only amplitudes, the unmodified attributes, such as centers and covariances, identify each primitive regardless of its storage order. The shared key uses these identities to reconstruct carrier order and dithers, so the decoder operates directly on the saved representation rather than on the rendered phase or holographic reconstruction. Clean and stego states share the same container schema, and a fixed-size protected descriptor supplies bounded recovery metadata without an external carrier manifest. Supplementary Secs. S3, S4, and S6 provide the complete baseline formulation, factorization and objective checks, and carrier-selection and recovery details.

3. EXPERIMENTAL SETUP

We use the framework of Kavakli et al. [1] to generate three 4094×2400-pixel single-color DPH frames at 639, 515, and 473 nm with a 3.74 μm pixel pitch. We evaluate the intended focal plane at $+2.5$ mm using band-limited angular-spectrum propagation [14]. The 100-bit benchmark uses one fixed centered 512×512 patch from each of 200 sources, comprising 100 CLIC 2020 [15] and 100 DIV2K [16] images. Before fitting the 150

newly selected sources, we fixed the carrier, QIM factor, primitive budgets, training schedules, serializers, and baseline adaptations. We prototype with Python 3.14.4, PyTorch 2.12.0+cu130, Odak [17], and an NVIDIA GeForce RTX 5090.

Baselines and evaluations. We compare **GS** with **GaussianImage**, a per-case **INRSteg** adaptation [10], and the released 100-bit **RoSteALS** checkpoint [11]. The methods differ in what their decoders read. **GaussianImage** and **GS** use 16,000 primitives per checkerboard component and decode the payload from the saved primitive parameters, whereas **INRSteg** and **RoSteALS** decode it from their serialized stego phase. For **GaussianImage** and **GS**, equal primitive count does not imply equal storage or compute. For optical evaluation, we reinsert each candidate patch into the clean full hologram, propagate the complete phase pattern, and compare the same focused region with the clean reconstruction using PSNR, SSIM [18], and LPIPS [19]. We compare phase PSNR using the alignment method of Choi et al. [20]. See Supplementary Sec. S2 for details.

4. RESULTS

Clean representation. At equal primitive count, both methods use 16,000 primitives per component. Before embedding, **GS** exceeds **GaussianImage** by 7.64 dB in Phase Distribution Alignment PSNR (PDA-PSNR) and 5.12 dB in reconstruction PSNR over 200 sources. An equal-parameter comparison, counting the final stored representation, uses 16,000 primitives per component for **GS** and 24,000 for **GaussianImage**. **GS** exceeds **GaussianImage** by 8.06 dB in PDA-PSNR and 5.44 dB in reconstruction PSNR.

Fixed-payload comparison. At 100 bits, **GS** gives the highest mean PDA-PSNR and reconstruction PSNR and the lowest LPIPS-Alex, see Table 1. **GS** exceeds **RoSteALS** in reconstruction PSNR by 2.23 dB (95% confidence interval [1.98, 2.49] dB) and is higher for 173 of 200 sources, whereas their mean SSIM differs by only 0.001. Both **GaussianImage** configurations, **GS**, and **INRSteg** recover all payloads from their respective decoder inputs, while **RoSteALS** incurs a single bit error across embedded bits. In the equal-parameter comparison, **GS** exceeds **GaussianImage** by 5.86 dB in PDA-PSNR and 5.44 dB in reconstruction PSNR over the same 200 sources.

Table 1. Mean Quality at 100 Bits over 200 Sources

		Holographic reconstruction		
Method	PDA-PSNR ↑	PSNR ↑	SSIM ↑	LPIPS ↓
INRSteg	24.44	20.17	0.325	0.759
GaussianImage-16k	40.17	37.89	0.917	0.138
GaussianImage-24k	39.78	37.57	0.910	0.143
RoSteALS	40.80	40.78	0.973	0.048
GS (Ours)	**45.64**	**43.01**	**0.974**	**0.029**

Explicit-representation settings

Method	Primitives/component	Parameters/patch
GaussianImage-16k	16k	256k
GaussianImage-24k	24k	384k
GS (Ours)	16k	384k

Counts include all stored parameters, including two frequency indices per **GS** primitive. PDA-PSNR evaluates aligned phase. The remaining metrics evaluate the reconstruction. PSNR is in dB. LPIPS denotes LPIPS-Alex.

Hardware evaluation. Fig. 2 shows comparable scene structure and local detail in **GS** captures before and after 100-bit embedding under identical settings.

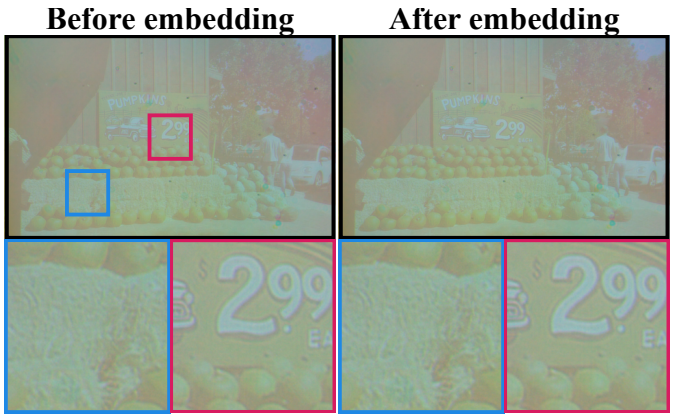


Fig. 2. Hardware evaluation. Full-frame captures before and after 100-bit embedding. Colors link the matched crops below.

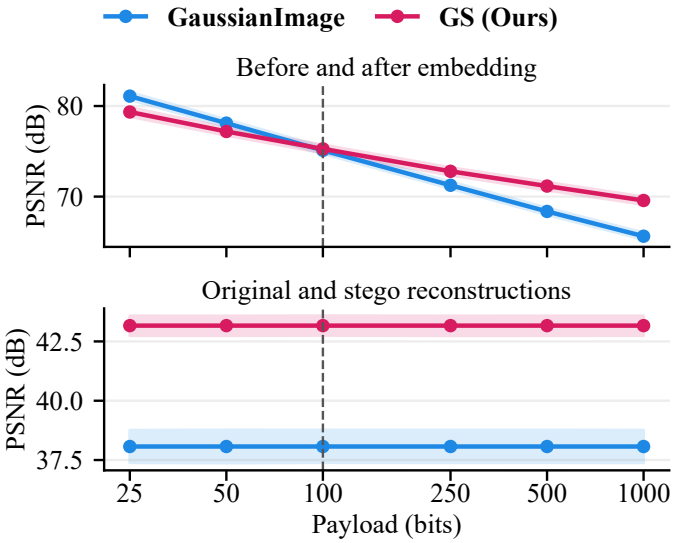


Fig. 3. Payload-size effect over 100 CLIC sources. The upper panel compares reconstructions before and after embedding, while the lower compares original-DPH and stego reconstructions. Shading shows 95% source-level intervals.

Payload size. We evaluate 25–1000 bits by comparing reconstructions before and after QIM embedding and by comparing the original-DPH and stego reconstructions. The first comparison isolates embedding distortion. The second includes fitting and embedding distortion. On 100 CLIC validation sources, the first PSNR for **GS** exceeds the second by 26.40–36.17 dB, and the second changes by less than 0.001 dB across the range (Fig. 3). **GaussianImage** adds less embedding distortion at small payloads, but **GS** overtakes it near 100 bits on CLIC and at 250 bits on DIV2K. Both decoders recover all 4,500 embedded payloads after save and reload. Figure 4 compares the stored phase and reconstructed image for CLIC IMG_1115.

Detectability and robustness. Without the shared key, classifiers using summaries of the saved primitive parameters or SRNet [21] on the stored phase yield ROC-AUC 0.500 at 100 and 1,000 bits. Recovery also remains exact after primitive reordering and reduced-precision storage. Supplementary Secs. S4, S5, and S7 give the complete protocols and results.

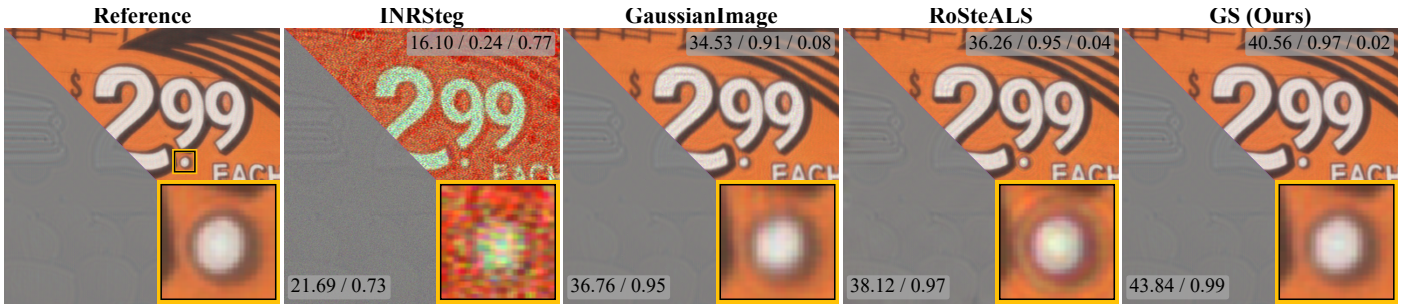


Fig. 4. Illustrative 100-bit comparison. GaussianImage and **GS** use 16,000 primitives per component. Diagonal panels show reconstruction and stored phase. Insets enlarge the same 40×40 reconstruction crop. Annotations report reconstruction PSNR/SSIM/LPIPS and PDA-PSNR/PDA-SSIM.

5. DISCUSSION

Stego reconstruction error combines representation fitting and QIM distortion. In Fig. 3, PSNR before and after embedding is 26.40–36.17 dB higher than PSNR between the original-DPH and stego reconstructions at every payload, showing that fitting dominates the final error. The equal-parameter comparison confirms **GS**'s fidelity advantage over **GaussianImage**. **GS** and **RoSteALS** have nearly equal SSIM despite their PSNR gap, so no single metric fully describes stego fidelity. Because the methods decode the payload from different inputs, their bit-error rates are not directly comparable.

Limitations. Our experiments use single-plane holograms. Future work should relate payload capacity to the number of reconstructed planes while preserving quality at every depth. The comparison matches parameters, not runtime, because the current **GS** implementation is substantially slower than **GaussianImage**.

Conclusion. This work establishes that a DPH can hide information in the parameters of its saved explicit representation, not only in its phase samples or generation process. Because these representations support hologram compression and streaming [8], payloads can travel with holographic content. Future work should explore emerging representations such as Beta Splatting [22] and extend the method to random-phase [23] and multi-color holograms [1].

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AI use. We used OpenAI ChatGPT and Codex (GPT-5-based models) for literature review, language editing, and coding assistance. We verified all outputs.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data and code will be released on GitHub upon acceptance.

Supplemental document. See Supplement 1 for details.

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